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Telecom Report

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General Introduction

In the modern era, DSL, telephone, and fiber optics are vital technologies that have transformed communication and connectivity. DSL provides high-speed internet access, while the telephone enables real-time conversations and meaningful connections. Fiber optics serves as the backbone, powering the internet and supporting bandwidth-intensive applications. Together, these technologies drive innovation, economic growth, and social connectivity in our interconnected society.

Chapter 1 lays the groundwork by presenting fundamental concepts related to telephony. This section provides a solid introduction to the subject matter, ensuring a common understanding of the key principles.

Moving forward, Chapter 2 delves into the crucial role of fiber optics. It delves into the various modes of fiber optics and explores the different types of connections used. This chapter sheds light on the significance of fiber optics in facilitating high-speed communication.

Chapter 3 focuses on the construction of central infrastructure for fixed telephone lines. It outlines a step-by-step approach to building a central facility, emphasizing the components involved and the methods employed to connect cables for end users. This section provides valuable insights into the physical aspects of establishing fixed telephone networks.

Lastly, Chapter 4 delves into the world of DSL (Digital Subscriber Line) technology. It examines the different types of DSL and how it connects to end users, providing them with reliable internet access over existing telephone lines.

Chapter 1: Basics

Before going into the upcoming chapters, it is important to establish some foundational information:

The equation $R = \rho L / S$ is a key aspect when considering transmission over copper wires. Here, R represents resistance, ρ denotes the resistivity of the material (copper), L signifies the length of the wire, and S represents the cross sectional area of the wire. It is worth noting that according to this equation, $R \propto 1/S$.

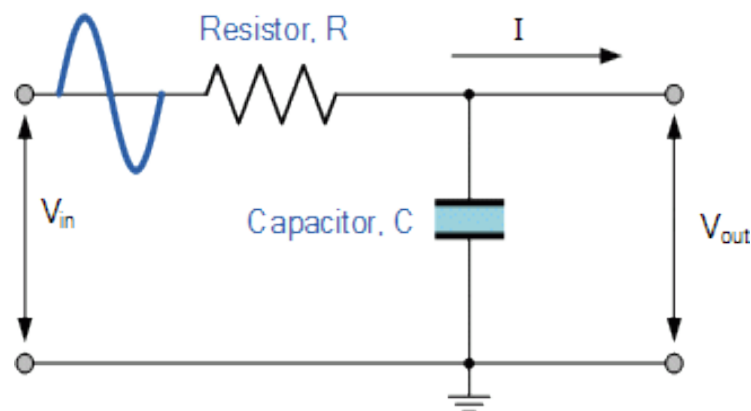
Telecom devices predominantly utilize direct current DC.

A rectifier is a device that converts AC signal into DC signal.

Each telecom central incorporates rectifiers connected to both load and batteries. The presence of batteries is essential to maintain connection services in the event of AC power failure. The batteries are connected in series, with specific current and cycle requirements.

Air conditioning plays a crucial role in cooling the devices in a central and preventing overheating.

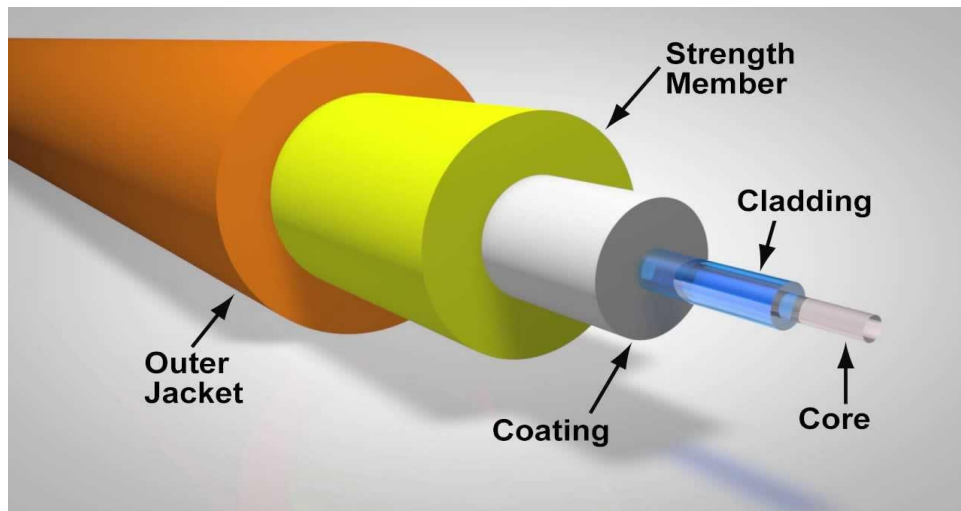
A low pass filter is used in DSL splitters, it generally consists of a capacitor and a resistor connected as shown below:



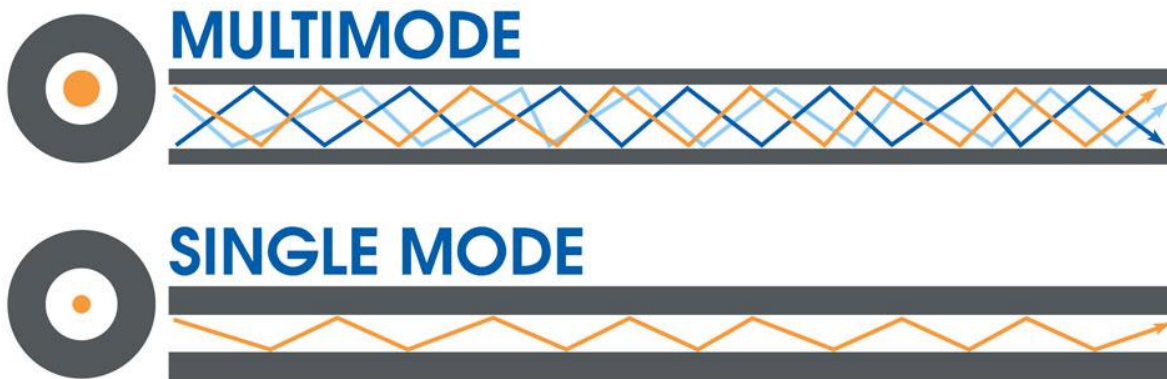
Chapter 2: Fiber Optics

Fiber optics is a technology that enables the transmission of data using light.

The fiber consists of 4 layers including the core, cladding, buffer and the jacket.



The applications of fiber optics are vast, particularly in long-distance and high-performance data networking. They are widely used in telecommunication services such as internet, television, and telephones. Two types of fiber optic cables are single-mode and multi-mode fibers, each suited for different distance requirements due to variations in core diameter. Single mode has 8-10 micro meters core diameter, used for long distances and sourced from laser. Multimode has larger diameter and can transmit multiple signals at once; it is sourced from LED.



An essential component of fiber optic cables is the fiber optic physical connector, which comes in different types, we will discuss 2 types depending on the shape of the connector. APC (angled physical contact) and UPC (ultra physical contact) connectors.

APC connectors are better than UPC in general because they have better performance.

To differentiate between fiber wires, colors are used. Each of the twelve colors, blue, orange, green, brown, grey, white, red, black, yellow, purple, pink, and turquoise, represents a specific fiber color and corresponds to the color of tubes containing twelve fibers each. This color coding system allows for the identification of different fiber cables, particularly in larger installations with multiple fibers.

Fiber optic cables offer several advantages over traditional copper cables. Firstly, they have higher bandwidth capacities, allowing for faster and more efficient data transmission.

Additionally, fiber optic cables are also less sensitive to interference, making them more reliable in environments with electromagnetic interference that include places with a lot of wires and motors around.

Fiber optic cables are stronger, thinner, and lighter compared to copper wires. They require less maintenance and replacement, providing long-lasting performance.

Chapter 3: Telephone Lines

How to build a central:

- 1- Get the map that shows all buildings and roads in the target area.
- 2- Visit the area and check for the information in person, and then update the map.
- 3- Canvassing and estimate the number of lines needed.
- 4- The central should be built in the area that is most populated.

The central consist of commercial area and technical area. In the technical area there are components such as the TDM (time division multiplexer), MDF (main distribution frame), batteries, generator, rectifier, solar, UPS and EDL.

To connect the user with the central, we create a cabinet called CCP and we connect the central to it using primary cable, then from the CCP to the end user.

A central consist of elements such as MDF (main distribution frame), TDM (time division multiplexer), generator, rectifier, batteries, solar panels to provide electricity.

The MDF consist of horizontal and vertical lines, the primary cable is terminated in a place called cable vault to make the cables VRB resistant to fire. It is then connected to the vertical, and the TDM wires connected to the horizontal, the TDM is responsible of certain sound signals in the telephone.

Wires have different colors: blue, orange, green, brown, grey.

They can be paired with white wire to make them 5 pairs. Another 5 paired with red colored wire. Each 10 pairs are wrapped with a colored ribbon.

Ribbon colors are: blue, orange, green, brown, grey, white, red, black, yellow and violet.

Each 100 pairs are wrapped with a white ribbon that has a number on it.

We can connect cables either underground or on electric poles. Usually primary cable connected to the CCP is buried while secondary cables are on electric poles.

If the cable is not enough length we create manhole/handhole with joints to join cables together.

Steps to bury primary cable underground:

- 1- Cut section of the road.

- 2- Add cement support for pipes.
- 3- Place the pipes on the support.
- 4- Add sand or powdered stone.
- 5- Test using metal ball and string.

Steps to connect secondary on poles:

- 1- Use wooden pole around 8 meters.
- 2- Peel the pole.
- 3- Dry the wood using an oven (500 degreeC).
- 4- Writing on the wood.
- 5- Use the oven again but while placing it in tar.
- 6- Use rocks and soil to place the pole.
- 7- Use steel backed wires that can handle heat.



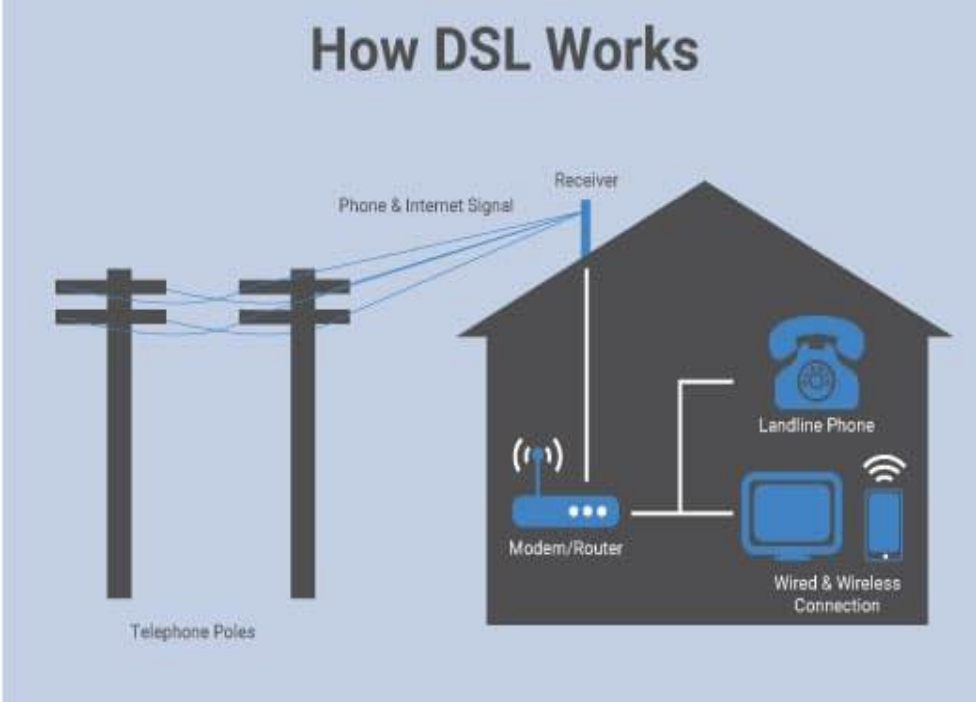
Chapter 4: DSL

DSL stands for Digital Subscriber Line and it is a way to transmit and receive data through copper telephone landline. DSL uses telephone cables to transmit data using a DSL modem allowing users to access internet.

xDSL is a reference to different DSL types, these types include ADSL, VDSL, SDSL and MDSL. The latter 2 are not used in Ogero. ADSL stands for Asymmetric DSL in which download and upload speeds are different. SDSL is Symmetric DSL where both download and upload speeds are equal. VDSL is very high speed DSL only available to users close to the central (around 600m away).

Users connect through the modem through telephone lines to a device called DSLAM (Digital Subscriber Line Access Multiplexer), this device consists of port panel and management cards, the role of the management card is to manage the transmission of data, and storing data so if any fault occurs in the system, it automatically switches to the other functioning management card without any losses. In case all management cards are active, the downside will become its uplink's transmission speed.

Ogero has its own method of giving people DSP access to users far off the central (around 2.5 km off). The method is about uniting with local authority to create a 3x3x3 room which has a panel of splitters, DSLAM, batteries and rectifier...using a fiber cable from the central connects to the uplink, and the voice off the main cable, the data will get to reach more far areas without loss.



General Conclusion

This report discussed very important topics in the world of telecom, being fiber optics, telephone landline and DSL. This knowledge will help me further develop my skills in telecom as I thrive to gain real time work experience, away from theorizing in regular classrooms.

Learning about DSL and landline have helped me gain more insights on what most people use and is essential to our modern life though I never knew how it works.

The world of telecom is always an active and developing field with more breakthroughs giving people faster and more reliable ways of communication. Science have managed to reach what have been called impossible speeds in the previous decade. Though this poses a question of the new possibilities, we could use this advancement for.

